

ZHYMPITY, KazakhstanWMO: **352170**Lat: **50.250N**Lon: **52.567E**Elev: **32**StdP: **100.94**Time Zone: **5.00 (E05)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-27.6	-24.3	-30.5	0.2	-27.5	-27.1	0.3	-24.0	14.4	-3.1	12.3	-6.1	2.2	20	0.590

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.9	36.5	19.9	34.7	19.3	32.9	18.7	21.4	32.8	20.5	31.5	19.6	30.2	3.9	140

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
17.9	12.9	24.7	16.7	11.9	24.5	15.5	11.0	24.3	62.1	32.8	58.9	31.5	56.1	30.2	28.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4)	11.6	9.9	8.4	-30.3	38.9	4.1	1.9	-33.3	40.2	-35.7	41.3	-38.0	42.4	-41.0	43.8	
(5)				-30.4	24.0	4.1	1.6	-33.4	25.1	-35.8	26.0	-38.1	26.9	-41.0	28.0	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	7.2	-10.9	-10.5	-2.6	9.1	16.8	21.8	24.1	22.9	15.7	7.3	-1.1	-7.8
	DBStd	13.84	6.76	7.11	5.94	5.59	4.82	4.59	3.62	4.39	4.75	4.73	5.91	6.63
	HDD10.0	2701	647	574	390	82	5	0	0	0	8	112	333	551
	HDD18.3	4597	905	807	648	281	87	18	2	11	104	342	582	809
	CDD10.0	1671	0	0	1	54	217	354	436	401	180	28	1	0
	CDD18.3	523	0	0	0	2	41	122	179	153	27	0	0	0
	CDH23.3	6623	0	0	0	37	537	1587	2221	1890	347	5	0	0
	CDH26.7	3141	0	0	0	4	198	781	1099	942	116	1	0	0
Wind	WSAvg	3.7	4.0	4.1	4.2	4.0	3.7	3.5	3.3	3.2	3.3	3.5	3.6	3.8
Precipitation	PrecAvg	259	18	15	18	21	27	28	28	15	19	27	24	19
	PrecMax	485	42	37	41	49	79	84	146	58	107	68	50	61
	PrecMin	134	4	2	0	0	3	4	1	1	1	4	1	4
	PrecStd	84	9	9	13	13	22	20	34	16	22	18	13	12
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	2.0	1.9	16.1	27.2	33.2	38.1	38.1	38.2	33.2	23.5	13.2	4.9
		MCWB	1.1	0.6	9.5	16.3	17.9	21.0	20.3	20.3	16.7	13.5	10.4	3.5
	2%	DB	0.7	1.0	10.7	23.9	31.0	36.0	36.1	36.1	29.8	20.8	9.9	2.4
		MCWB	-0.1	0.1	7.1	14.8	17.5	19.9	19.8	19.8	16.6	12.2	7.8	1.4
	5%	DB	0.1	0.3	7.1	21.3	29.0	33.6	34.4	34.0	27.4	18.0	7.5	1.1
		MCWB	-0.6	-0.5	4.6	13.5	17.0	19.0	19.4	18.9	15.6	10.6	5.8	0.3
	10%	DB	-1.3	-0.7	4.4	18.4	26.4	31.5	32.4	31.9	24.7	15.2	5.8	0.3
		MCWB	-2.1	-1.6	2.4	11.7	16.1	18.4	18.9	18.3	14.6	9.5	4.2	-0.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	1.2	1.2	9.7	17.4	20.2	23.0	22.5	22.4	19.3	15.0	11.3	4.0
		MCDB	1.6	1.4	14.7	24.9	28.9	32.8	32.9	33.7	30.1	21.9	12.7	4.6
	2%	WB	0.2	0.4	7.3	15.5	18.6	21.2	21.2	21.0	17.4	12.8	8.0	1.7
		MCDB	0.6	0.8	10.8	23.1	28.4	33.0	32.1	33.2	27.2	18.7	9.7	2.2
	5%	WB	-0.6	-0.5	4.8	13.8	17.6	20.0	20.5	19.9	16.2	11.5	6.1	0.5
		MCDB	0.0	0.2	7.1	20.1	27.2	31.2	31.0	31.5	25.6	16.6	7.5	1.0
	10%	WB	-2.1	-1.6	2.6	12.1	16.5	19.0	19.7	19.0	15.1	10.1	4.4	-0.5
		MCDB	-1.2	-0.7	4.1	17.8	25.6	29.3	29.8	30.1	23.6	14.3	5.7	0.2
Mean Daily Temperature Range	5% DB	MDBR	6.9	7.8	7.8	11.4	12.5	13.3	12.9	13.4	13.0	10.6	6.4	6.1
		MCDBR	4.8	5.8	10.9	14.7	15.4	15.9	15.1	15.4	16.7	15.2	8.0	4.7
		MCWBR	4.8	5.4	7.7	7.9	6.7	6.4	5.6	5.7	7.6	8.5	6.3	4.4
	5% WB	MCDBR	4.7	5.4	10.3	13.9	14.4	14.2	13.3	14.1	14.9	13.3	7.2	4.6
		MCWBR	4.9	5.3	7.5	8.0	7.3	6.8	6.1	6.3	7.5	8.4	6.0	4.4
Clear-Sky Solar Irradiance	taub	0.317	0.339	0.394	0.444	0.425	0.414	0.416	0.428	0.380	0.361	0.342	0.316	
	taud	2.066	2.012	2.032	2.050	2.186	2.283	2.287	2.225	2.320	2.293	2.223	2.127	
	Ebn at Noon	671	762	788	795	833	845	837	804	807	745	650	613	
	Edh at Noon	85	117	138	152	139	127	125	127	103	88	74	70	
All-Sky Solar Radiation	RadAvg	1.16	2.28	3.39	4.66	6.04	6.63	6.43	5.55	4.00	2.26	1.04	0.73	
	RadStd	0.21	0.26	0.49	0.39	0.47	0.51	0.38	0.34	0.36	0.27	0.13	0.19	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (3 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page